CSA0509 – DATABASE MANAGEMENT SYSTEM

CO1-AT1

SHORT ANSWERS

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QUESTION.1

1. Explain Schema and Instance with Examples. Distinguish between Logical Schema, Physical Schema, and View Schema.

Schema

A **schema** is the overall design or blueprint of a database. It defines the structure of the database, including tables, attributes, relationships, constraints, and data types.

Example:

StudentID	Name	Department	Age

Student Table Schema

CREATE TABLE Student(

StudentID INT PRIMARY KEY,

Name VARCHAR(50),

Department VARCHAR(30),

Age INT

);

This structure is called the **schema**.

Instance

An **instance** is the actual data stored in the database at a particular point in time.

Example

StudentID	Name	Department	Age
101	**Anu**	**AI & DS**	**20**
102	**Anitha**	**CSE**	**21**
103	**Priya**	**ECE**	**20**

The data may change every day, but the schema usually remains unchanged.

Difference between Schema and Instance

Schema	Instance
Blueprint of database	Actual data stored
Rarely changes	Changes frequently
Defines structure	Contains records
Created by DBA	Updated by users

Types of Schema

1. Logical Schema

- Describes the logical structure of the database.
- Includes tables, attributes, relationships, and constraints.
- Independent of hardware.

Example:

Student(StudentID, Name, Department, Age)

Course(CourseID, CourseName)

2. Physical Schema

- Describes how data is physically stored.
- Includes storage blocks, indexing, file organization.

Example:

Student table stored in Block 20

B+ Tree Index on StudentID

3. View Schema

- Shows only a portion of the database.
- Different users see different views.

Example

Faculty View

StudentID	Name	Department

Student View

StudentID	Name	Age

QUESTION.2

Identify the entities, attributes, and relationships for a Bank Management System and construct its ER diagram.

Entities

Customer

Attributes

- Customer_ID (PK)
- Name
- Address
- Phone
- Email

Account

Attributes

- Account_No (PK)
- Account_Type
- Balance
- Opening_Date

Branch

Attributes

- Branch_ID (PK)
- Branch_Name
- Location

Loan

Attributes

- Loan_ID (PK)
- Loan_Amount
- Interest_Rate

Employee

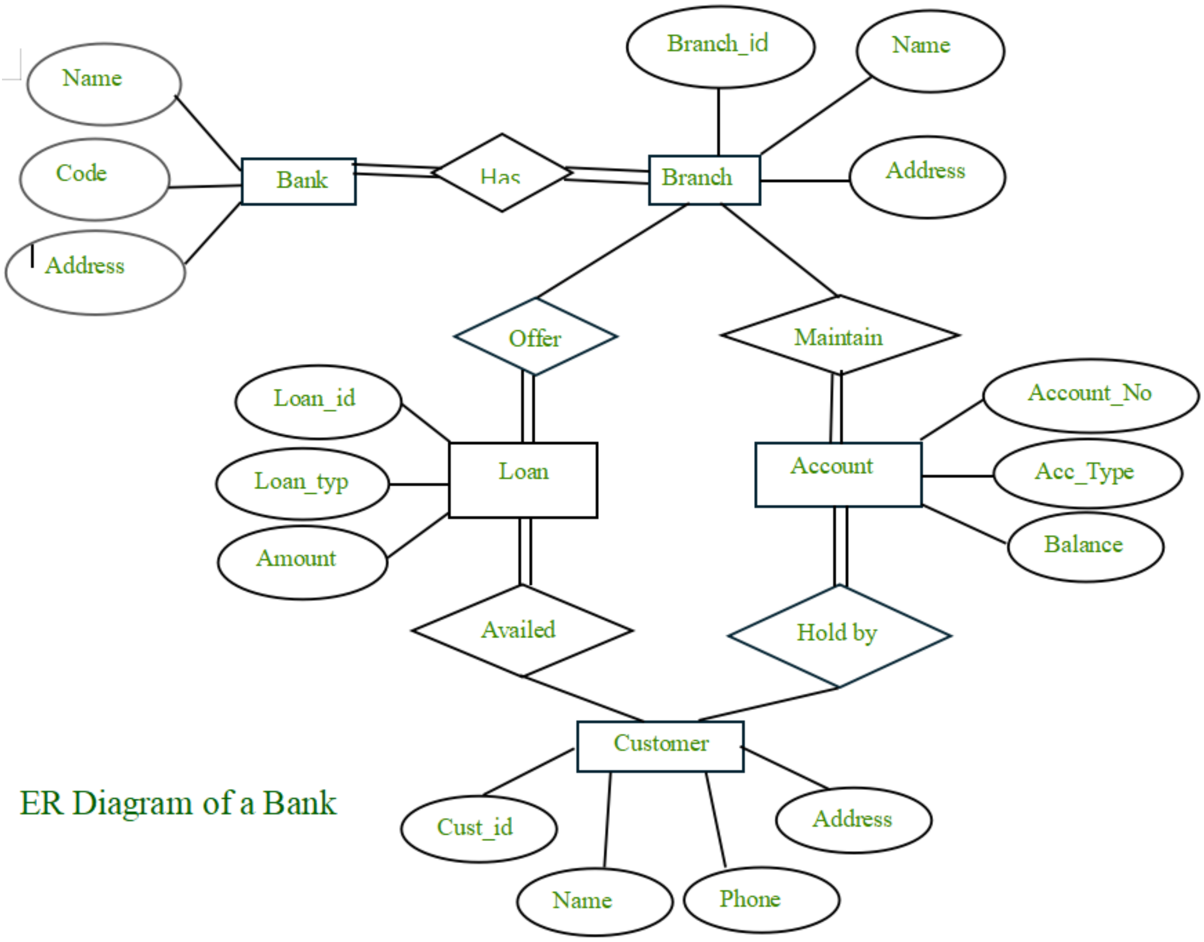
Attributes

- Employee_ID (PK)

- Employee_Name
- Designation
- Salary

Relationships

- Customer owns Account
- Branch maintains Account
- Customer takes Loan
- Branch provides Loan
- Branch employs Employee



ER Diagram of a Bank

QUESTION.3

Explain the Extended ER (EER) Model. Describe the concepts of Generalization, Specialization, and Aggregation.

Extended ER (EER) Model

The Extended Entity Relationship (EER) model is an advanced version of the ER model. It supports inheritance, subclasses, superclass, specialization, generalization, and aggregation.

Features:

- Superclass and Subclass
- Inheritance
- Generalization
- Specialization
- Aggregation

Generalization

Generalization combines multiple lower-level entities into a higher-level entity.

Car Bike Truck

Vehicle

Manager and Clerk inherit Employee attributes.

Specialization

Specialization divides one higher-level entity into smaller specialized entities.

Employee

Manager Clerk

Manager and Clerk inherit Employee attributes.

Aggregation

Aggregation treats a relationship as a higher-level entity.

Example

Employee ---- Works_On ---- Project

Aggregated as Assignment

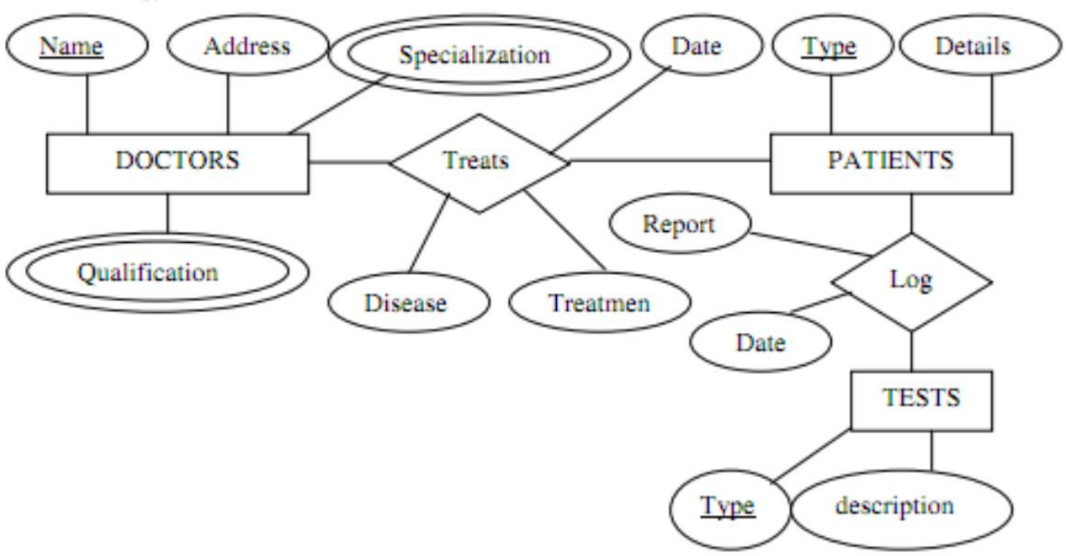
This allows another entity to participate in the relationship.

Comparison:

Generalization	Specialization
Bottom-Up	Top-Down
Combines lower-level entities into a higher-level entity	Divides a higher-level entity into lower-level entities
Creates a superclass	Creates subclasses
Example: Vehicle from Car and Bike	Example: Employee into Manager and Clerk

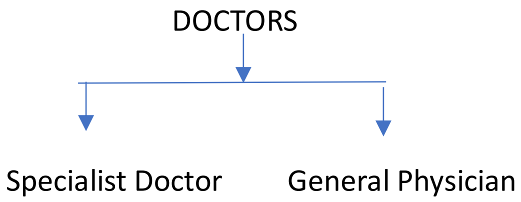
QUESTION:4

Design an EER Diagram for a Hospital Management System incorporating inheritance and weak entities.



Inheritance (Specialization):

The DOCTORS entity can be specialized into:



Both subclasses inherit:

- Doctor_ID
- Name
- Address
- Qualification

The Specialist Doctor additionally uses the Specialization attribute.

Weak Entity:

The TESTS entity is considered a Weak Entity because:

- It depends on the PATIENTS entity.
- A test record cannot exist without a patient.
- It is identified using the Patient along with the Test_ID (Partial Key).

Conclusion

The Hospital Management System EER model effectively represents the relationships among Doctors, Patients, and Tests. It uses inheritance to classify doctors into specialized categories and treats Tests as a weak entity because they cannot exist independently without a patient. This design improves data organization, reduces redundancy, and supports efficient hospital database management.

QUESTION.5

Explain the role of a DBA (Database Administrator) and the responsibilities associated with the position.

Definition

A Database Administrator (DBA) is responsible for installing, configuring, maintaining, securing, and managing databases to ensure efficient performance and data availability.

Responsibilities

1. Database Installation

- Install DBMS software.
- Configure database servers.

2. Database Design

- Create tables.
- Define relationships.
- Create indexes.

3. Security Management

- Create user accounts.
- Grant and revoke privileges.
- Protect data from unauthorized access.

4. Backup and Recovery

- Schedule backups.
- Restore lost data.
- Implement disaster recovery plans.

5. Performance Tuning

- Optimize SQL queries.

- Create indexes.
- Improve response time.

6. Storage Management

- Allocate storage.
- Monitor disk usage.
- Manage files.

7. Data Integrity

- Enforce constraints.
- Maintain consistency.
- Prevent duplicate records.

8. Database Monitoring

- Monitor server performance.
- Detect errors.
- Ensure high availability.

9. User Support

- Assist developers.
- Resolve database issues.
- Maintain documentation.

Skills Required

- SQL
- Database Design
- Backup & Recovery
- Security Management
- Performance Tuning

- Problem Solving

Advantages of a DBA

- Ensures data security.
- Maintains database performance.
- Prevents data loss.
- Supports multiple users.
- Ensures high availability and reliability.